

Foundation (Jalapeno)

Q1. What is the vertex of the parabola $y = x^2 + 4x + 4$?

- (A) (0,0)
 (B) (2,0)
 (C) (-2,0)
 (D) (-2,4)
 (E) (-2,-4)

(A) 0

(B) 1

(C) -1

(D) 2

(E) 4

Q2. What is the axis of symmetry for $y = x^2 - 6x + 8$?

- (A) $x = 2$
 (B) $x = -2$
 (C) $x = 3$
 (D) $x = 1$
 (E) $x = 6$

Q6. What is the vertex form of $y = x^2 + 4x + 4$?

- (A) $y = (x + 2)^2$
 (B) $y = (x - 2)^2$
 (C) $y = (x + 2)^2 + 4$
 (D) $y = (x - 2)^2 + 1$
 (E) $y = (x + 1)^2$

Q3. What are the y -intercepts of the parabola $y = (x+2)(x-3)$?

- (A) $y = -6$
 (B) $y = 6$
 (C) $y = -2$ and $y = 3$
 (D) $y = 2$ and $y = -3$
 (E) $y = 0$

Q7. For the parabola $y = 2x^2 - 4x - 1$, the x -coordinate of the vertex is...

- (A) 0
 (B) 1
 (C) -1
 (D) 2
 (E)

Q4. The parabola $y = -x^2 + 2x + 3$ opens...

- (A) upward
 (B) downward
 (C) leftward
 (D) rightward
 (E) it is a straight line

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Q8. The equation of the axis of symmetry for $y = ax^2 + bx + c$ is...

- (A) $x = -\frac{b}{2a}$
 (B) $x = \frac{b}{2a}$
 (C) $y = -\frac{b}{2a}$
 (D) $y = \frac{b}{2a}$
 (E) $x = 0$

Q5. The minimum value of the function $y = x^2 + 2x + 1$ is..

Open Ended Questions (FRQ)

Q9. Graph the parabola $y = x^2 - 3x - 4$. Identify the vertex, axis of symmetry, and y -intercept.

Q10. Find the maximum or minimum value of $y = -x^2 + 4x - 3$ and the x -value at which it occurs.

Chef's Tips

- Tip 1: Check for understanding of vertex form of a parabola
- Tip 2: Ensure students can identify key features of a parabola from its equation
- Tip 3: Emphasize the importance of labeling graphs with key points and features